

SUDONG LEE

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EDUCATION

Ph.D. student in Robotics, Control, and Intelligent Systems (EDRS) <i>EPFL (École Polytechnique Fédérale de Lausanne)</i> <i>Advisor: Prof. Josie Hughes</i>	2023.04. - present <i>Lausanne, Switzerland.</i>
M.S. in Mechanical Engineering <i>Seoul National University</i> <i>Advisor: Prof. Yong-Lae Park</i> <i>Thesis: Modularized Robotic Skin Sensorized by Fiber Optic Force Sensing for Remote and Autonomous Robot Operation</i>	2019.03. - 2021.08. <i>Seoul, Korea.</i>
B.S. in Mechanical Engineering <i>Korea University</i>	2013.03. - 2019.02. <i>Seoul, Korea.</i>

RESEARCH EXPERIENCE

CREATE Lab (Computational design of Robots and Embodied Autonomous TEchnologies Laboratory) - <i>EPFL</i> Doctoral Assistant Research topics: · Morphological effects of finger bone length variation on manipulation · Robotic skin enabled by layered structures and soft sensing mechanisms · Robotic shoulder joint with compliant mechanism and passive stabilization	<i>Lausanne, Switzerland.</i> 2023.04. - present
Soft Robotics Research Center (SRRC) - <i>Seoul National University</i> Research Associate Research Assistant Research topics: · Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception · Soft electronics and sensors using selective patterning of conductive Elastomers	<i>Seoul, Korea.</i> 2022.09. - 2023.03. 2021.09. - 2022.08.
Soft Robotics and Bionics Laboratory (SRBL) - <i>Mechanical Engineering, Seoul National University</i> Graduate Student Researcher Research topics: · Multi-funtional soft sensor enabled by heterogeneous sensing mechanisms · Robotic skin sensorized with fiber bragg gratings (FBGs) · Environmental adaptability of legged robots with cutaneous inflation and sensation	<i>Seoul, Korea.</i> 2019.01. - 2021.08.

HONORS AND AWARDS

M.S. Thesis Presentation Award Mechanical Engineering, Seoul National University	2021.06.
Third Place Award for Locomotion Challenge IEEE International Conference on Soft Robotics 2019 (RoboSoft 2019) Team SRBL (Sudong Lee, G. Shin, J. Kim, M. Choi, Y. Baek, and Y.-L. Park)	2019.04.
Great Honor, Winter 2018 Graduation Korea University	2019. 02.
Semester High Honors Korea University	1 st Semester, 2018., 2 nd Semester, 2017., 1 st Semester, 2015., 2 nd Semester, 2014., 1 st Semester, 2014., 2 nd Semester, 2013., 1 st Semester, 2013.

SCHOLARSHIPS

Kwanjeong Fellowship Kwanjeong Educational Foundation	2 nd Semester, 2020., 1 st Semester, 2020., 2 nd Semester, 2019., 1 st Semester, 2019.
National Science and Engineering Scholarship Korea Student Aid Foundation	2 nd Semester, 2018., 1 st Semester, 2018., 2 nd Semester, 2017., 1 st Semester, 2015.
Academic Excellence Scholarship Korea University	2 nd Semester, 2014.
Best Honor Scholarship Korea University	1 st Semester, 2014.

PUBLICATIONS

Journal Papers

1. **Sudong Lee** and J. Hughes, "Multi-modal, multi-contact, and scalable soft robot skin enabled by layered structures." (*Under review*)
2. **Sudong Lee**, S.M.H. Sadati, A. Georgopoulou, H. Hauser, F. Vollrath, and J. Hughes, "Spider-Web Inspired 3D Printed Soft Sensor Designs for Physical Information Structuring and Damage Resilience." (*Submitted*)
3. **Sudong Lee**, K. Gilday, and J. Hughes, "Tendon-driven robotic shoulder joint with passive stabilization for large workspace," *IEEE/ASME Transactions on Mechatronics*, 2026. (DOI: [10.1109/TMECH.2026.3674717](https://doi.org/10.1109/TMECH.2026.3674717))
4. A. Georgopoulou, M. A. Calvo, L. Lucherini, **Sudong Lee**, J. Hughes, and E. Amstad, "Programmable somatosensory soft robots," *npj Flexible Electronics*, vol. 10, 58, 2026. (DOI: [10.1038/s41528-026-00558-0](https://doi.org/10.1038/s41528-026-00558-0))
5. A. Georgopoulou, **Sudong Lee**, B. Dai, F. Bono, J. Hughes, and E. Amstad, "3D printing of self-healing longevous multi-sensory e-skin," *Communications Materials*, vol. 6, 121, 2025. (DOI: [10.1038/s43246-025-00839-7](https://doi.org/10.1038/s43246-025-00839-7))
6. **Sudong Lee***, J. I. Kim*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *IEEE Transactions on Robotics*, vol. 40, pp. 2373-2389, 2024. (DOI: [10.1109/TRO.2024.3378178](https://doi.org/10.1109/TRO.2024.3378178))
*: **Sudong Lee and J. I. Kim are co-first authors.**
7. D. Kim, **Sudong Lee**, T. H. Hong, and Y.-L. Park, "Exploration-based model learning with self-attention for risk-sensitive robot control," *npj Robotics*, vol. 1, 7, 2023. (DOI: [10.1038/s44182-023-00006-5](https://doi.org/10.1038/s44182-023-00006-5))
8. J. Kang*, **Sudong Lee***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *IEEE Robotics and Automation Letters*, vol. 8, no. 11, pp. 7344-7351, 2023. (DOI: [10.1109/LRA.2023.3316075](https://doi.org/10.1109/LRA.2023.3316075))
*: **Sudong Lee and J. Kang are co-first authors.**
9. T. Kim*, **Sudong Lee***, S. Chang, S. Hwang, and Y.-L. Park, "Environmental adaptability of legged robots with cutaneous inflation and sensation," *Advanced Intelligent Systems*, vol. 5, no. 11, 2300172, 2023. (DOI: [10.1002/aisy.202300172](https://doi.org/10.1002/aisy.202300172))
*: **Sudong Lee and T. Kim are co-first authors.**
Cover Article: [10.1002/aisy.202370050](https://doi.org/10.1002/aisy.202370050), Editor's Choice: [\[Link\]](#)
10. Y. Lee, S. Lim, W. J. Song, **Sudong Lee**, S. J. Yoon, J.-M. Park, M.-G. Lee, Y.-L. Park, and J.-Y. Sun, "Triboresistive touch sensing: grid-free touch-point recognition based on monolayered ionic power generators," *Advanced Materials*, vol. 34, no. 19, 2108586, 2022. (DOI: [10.1002/adma.202108586](https://doi.org/10.1002/adma.202108586))
11. G. Shin*, **Sudong Lee***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4983-4990, 2022. (DOI: [10.1109/LRA.2022.3153707](https://doi.org/10.1109/LRA.2022.3153707))
*: **Sudong Lee and G. Shin are co-first authors.**
12. T. Kim, **Sudong Lee**, T. Hong, G. Shin, T. Kim, and Y.-L. Park, "Heterogeneous sensing in a multifunctional soft sensor for human-robot interfaces," *Science Robotics*, vol. 5, no. 49, eabc6878, 2020. (DOI: [10.1126/scirobotics.abc6878](https://doi.org/10.1126/scirobotics.abc6878))

Conference Papers

1. Q. Fu, **Sudong Lee**, A. Grillo, J. Arreguit, L. Gevers, J. Hughes, and A. J. Ijspeert, "Polymander II: an amphibious salamander-inspired robot with contact and flow sensors," *2026 IEEE International Conference on Robotics and Automation (ICRA)*, Vienna, Austria, 2026. (Accepted)
*: **Sudong Lee and Q. Fu are co-first authors.**
2. **Sudong Lee** and J. Hughes, "Morphological and material programability of a hall-effect based soft tactile sensors," *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, San Diego, CA, USA, 2024, pp. 325-331. (DOI: [10.1109/RoboSoft60065.2024.10521990](https://doi.org/10.1109/RoboSoft60065.2024.10521990))

CONFERENCE PRESENTATIONS

1. A. Georgopoulou, **Sudong Lee**, B. Dai, J. Hughes, and E. Amstad, "Multimodal selective sensory receptors for robotic e-skin," *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)*, Lausanne, Switzerland, 2025.
2. **Sudong Lee***, J. I. Kim*, Y. Baek, D. Chang, J. Lee, Y. S. Park, D. Lee, and Y.-L. Park, "Fiber-optic force sensing of modular robotic skin for remote and autonomous robot control," *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Abu Dhabi, UAE, 2024.
*: **Sudong Lee and J. I. Kim are co-first authors.**
3. J. Kang*, **Sudong Lee***, and Y.-L. Park, "Soft bending actuator with fiber-jamming variable stiffness and fiber-optic proprioception," *2024 IEEE International Conference on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024.
*: **Sudong Lee and J. Kang are co-first authors.**
4. G. Shin*, **Sudong Lee***, and Y.-L. Park, "Selective patterning of conductive elastomers embedded with silver powders and carbon nanotubes for stretchable electronics," *2022 IEEE 5th International Conference on Soft Robotics (RoboSoft)*, Edinburgh, United Kingdom, 2022.
*: **Sudong Lee and G. Shin are co-first authors.**

PATENTS

1. T. Kim, **Sudong Lee**, and Y.-L. Park, "Soft Sensor with Multi-Sensing Function," 2022.
(Korea Patent: 10-2384623)

TEACHING EXPERIENCE

Assistive Technology Challenge (ATC) - HACKAHEALTH, EPFL Project coach Supervisors: Prof. Auke J. Ijspeert, Prof. Silvestro Micera, Prof. Josie Hughes	Spring semester, 2025-26.
ME-320: Product development and engineering design - Mechanical Engineering, EPFL Teaching assistant, Instructor: Prof. Josie Hughes.	Autumn semester, 2025-26.
Assistive Technology Challenge (ATC) - HACKAHEALTH, EPFL Project coach Supervisors: Prof. Auke J. Ijspeert, Prof. Silvestro Micera, Prof. Josie Hughes	Spring semester, 2024-25.
ME-320: Product development and engineering design - Mechanical Engineering, EPFL Teaching assistant, Instructor: Prof. Josie Hughes.	Autumn semester, 2024-25.
Interdisciplinary robot competition - EPFL Team coach, Instructor: Prof. Auke J. Ijspeert.	Spring semester, 2023-24.
ME-320: Product development and engineering design - Mechanical Engineering, EPFL Teaching assistant, Instructor: Prof. Josie Hughes.	Autumn semester, 2023-24.
M2794.001700_001: Mechanical product design - Mechanical Engineering, Seoul National University Teaching assistant, Instructor: Prof. Yong-Lae Park.	1 st semester, 2019.

ACADEMIC SERVICE

Reviewer - Journal Papers

IEEE Transactions on Robotics (T-RO)
IEEE/ASME Transactions on Mechatronics (T-MECH)
IEEE Transactions on Industrial Electronics (TIE)
IEEE Robotics and Automation Letters (RA-L)
Nature Communications
Soft Robotics

Reviewer - Conference Papers

IEEE International Conference on Robotics and Automation (ICRA)
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
IEEE International Conference on Soft Robotics (RoboSoft)

TECHNICAL STRENGTHS (SKILLS)

Programming Languages	C++, Python, Matlab
Embedded System	Arduino, AVR ATmega, Single-Board Computer (SBC)
Software for System and Robots	ROS, Pybullet
Machine Learning	Pytorch, TensorFlow
Design and Simulation	3D Computer-Aided Design (CAD), Finite Element Analysis (FEA) Software
Fabrication	3D Printing (Additive Manufacturing), Silicone Fabrication

OTHER EXPERIENCE

Republic of Korea Air Force (ROKAF, Military Service) Staff Sergeant, Honorable discharge.	2015.08. - 2017.08.
KORAL (Korean Association in Lausanne) President (2025.09. - present)	2023 - present
KSEAS (Korean Scientist and Engineers Association in Switzerland) Operating Committee Member	2025.02. - present